

AYAN GUPTA

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EDUCATION

University of Michigan Ann-Arbor : B.S Engineering, Class of 2030

A-Levels : Mathematics A*, Further Mathematics A*, Physics A* & Chemistry A*

SAT : 1550 (790 Math, 760 English)

TECHNICAL PROJECTS

Computational Analysis of CERN Data — Independent Project | Jun–Jul 2025 | [GitHub](#)

- Analyzed 100,000 muon events from proton-proton collisions to identify Z-boson decays.
- Used Gaussian/Cauchy models and Monte Carlo simulations to study invariant-mass distributions.
- Built a PyTorch multilayer perceptron, achieving 73% classification accuracy.

Carnegie Mellon University Qatar Lifelines Hackathon — Finalist, 12th/80 | 2026 | [GitHub](#)

- Built a Random Forest model for 24-hour community water-demand forecasts; evaluated with MAE/MSE.
- Used Google OR-Tools to optimize delivery routes under capacity, fuel-range, and demand constraints.

Shell.ai Hackathon — Top 15% | 2025 | [GitHub](#)

- Built an XGBoost model to predict fuel properties from mixture data across 10 target properties.

EXPERIENCE

Electrical Engineering Intern — Ooredoo | Jul–Aug 2025

- Analyzed 4G/5G coverage with CovMo, maintaining KPIs above 95%; compared Okumura-Hata, MYRIAD, and ray-tracing models.
- Calculated network-site battery backup requirements and inspected power-redundancy systems at a Tier 3 data center.

AWARDS & LEADERSHIP

Carnegie Mellon University Qatar Pi Day Mathematics Competition — 4th Place | Jan–Mar 2025

- Advanced through two rounds of national mathematics competition solving problems in algebra, geometry, probability and math reasoning. Awarded a cash prize.

Al Rayyan International University Innovation Challenge — 3rd Place | Apr 2025

- Co-developed a piezoelectric kinetic-energy-harvesting sidewalk and thermal solutions for Qatar. Awarded a cash prize.

Engineering Club, Park House English School — Co-Founder & Head | Sep 2024–Present

- Lead engineering projects and teach MATLAB through hands-on experiments like an Arduino mobile rover.

TECHNICAL SKILLS

Languages: Python, MATLAB, Arduino/C++

Libraries and Tools: PyTorch, XGBoost, scikit-learn, Google OR-Tools, MANIM, Git